| **Measurement** | **BFS** | **A*** |
|---|---|---|
| Optional Moves | 30 | 30 |
| States Expanded | 181,393 | 7,550 |
| States discovered/generated | 181,440 | 11,096 |

Both BFS and A* found an optimal solution of 30 moves. However, A* explored significantly fewer states than BFS.

BFS explores states according to their distance from the starting state. It does not have information about which states are closer to the goal, so it may explore many states that do not lead toward the solution.

A* uses the priority value: f(n) = g(n) + h(n)

where g(n) is the number of moves taken from the starting state and h(n) is the Manhattan distance estimate from the current state to the goal.

The Manhattan distance helps A* estimate which puzzle states are more promising. Because A* uses both the cost so far and an estimate of the remaining distance, it can focus its search more toward the goal. This is why A* expanded 7,550 states compared with 181,393 states for BFS, while still finding the same 30-move optimal solution.